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Assessing sustainable performance of aquatic species using multiple footprints for comprehensive dietary advice

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ABSTRACT

Aquaculture has become an important source of food, providing an alternative to wild fisheries and terrestrial food. However, the environmental impacts of aquaculture can vary greatly, and it is essential to understand the relationship between the nutrition provided by aquatic species and their multiple environmental footprints for targeted dietary advice and sustainable development of aquaculture. In this study, we examined 49 aquatic species or taxa divided into eight groups, considering ten nutrients and five environmental impacts for Chinese aquaculture. The results showed that species varied widely in their nutritional values and environmental impacts, with mussel, razor clam, marine bass, flounder and snapper being the most recommended for sustainable performance (sustainability score $S_i > 1.60$). In contrast, tilapia performed poorly ($S_i = 0.11$) in both nutritional score and environmental impacts, thus should not be promoted in dietary advice. Results of spatial analysis showed that Qinghai ($S_i = 1.06$) and Tibet ($S_i = 0.75$) had the highest sustainable development level for freshwater aquaculture, while there were large differences in marine aquaculture among coastal provinces (the highest S_i in Zhejiang was 1.42, and the lowest S_i in Guangxi was 0.56) and the sustainable performance of all coastal provinces is reduced to varying degrees when freshwater aquaculture and marine aquaculture are considered comprehensively. Increasing the proportion of high sustainable performance species is an effective way to improve the sustainability score in most provinces, particularly coastal ones. We suggest that the proportion of species production should be adjusted, and the cultivation of species with high sustainable performance should be prioritized. This study provides an initial assessment of the sustainability of aquatic food by combining nutrition and multiple environmental impacts for Chinese aquaculture.

1. Introduction

Aquaculture is increasingly becoming an important source of seafood for the world and has the potential to continue to grow with the increasing wealth and population growth (Boyd et al., 2020; Fiorella et al., 2021). With the highly diverse species in aquaculture, produced under various conditions and production systems in marine and inland environments (Naylor et al., 2021; Dong et al., 2022), it not only provides animal protein but also contributes to meeting human nutritional goals. Seafood is a rich source of n-3 fatty acids, vitamin D, vitamin B12, selenium, iodine, iron, zinc, and phosphorus (Shepon et al., 2021), which are essential for maintaining good health. In many developing countries, deficiencies of calories, protein, and micro-nutrients are common, posing major public health concerns (Golden et al., 2016; Shepon et al., 2021). Such deficiencies can lead to an increased risk of mortality, growth retardation, cognitive deficits, and

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immune function reduction. Studies have shown that seafood contains many essential nutrients not found in red meat and that it can offer beneficial nutritional value and health effects. Therefore, seafood is recommended to be included in a sustainable diet (Scarborough et al., 2014; Hallström et al., 2019).

Aquaculture has become an increasingly important source of seafood, providing both food and important nutrients to people worldwide, particularly in areas with high rates of malnutrition (Shepon et al., 2021). However, the diversity of species and production systems used in aquaculture can lead to significant environmental impacts (Jiang et al., 2022), including greenhouse gas emissions, energy use, and water pollution, among others (Bergman et al., 2020; Gephart et al., 2021; Guzmán-Luna et al., 2021; Huang et al., 2020; Jiang et al., 2022; MacLeod et al., 2019). To promote sustainable and healthy diets, it is important to understand the trade-offs between the nutritional benefits of different aquatic food options and their environmental impacts (Hallström et al., 2019). Hallström et al. (2019) analyzed the combined climate and nutritional performance of seafood commonly consumed in Sweden, and suggested dietary changes towards increased consumption of more aquatic food choices where a correlation exists (e.g., pelagic species like sprat, herring and mackerel) would benefit both health and climate. Bianchi et al. (2022) assessed the nutrient density and greenhouse gas emissions of global fisheries, and their results showed that the highest nutrient benefit at the lowest emissions is achieved by consuming wild-caught small pelagic and salmonid species, and farmed bivalves like mussels and oysters. Shepon et al. (2021) expanded the dimensions and the environmental impacts of this study, they included not only greenhouse gas emissions but also habitat destruction, freshwater pollution, etc. Based on a nutrition-sensitive approach they found no single system offering a complete nutrition-environment win-win in Indonesia. However, the relationship between nutrition and environmental impacts of aquatic food is not well-understood in China - the largest aquaculture producer in the world, which supplied 58% and 59% of the global aquaculture volume and value, respectively in 2017 (Cao et al., 2015; FAO, 2018; Naylor et al., 2021). Chinese aquaculture industry produces a variety of marine and freshwater species, including fish, crustaceans, bivalves, and seaweed (Gui et al., 2018). Despite the importance of Chinese aquaculture industry, the environmental impacts of aquatic food production were often poorly assessed, particularly in the context of dietary studies (Li et al., 2011). To promote sustainable aquaculture and healthy diets in China, it is essential to comprehensively evaluate the relationship between the nutrition provided by aquatic species and their multiple environmental footprints, considering factors such as greenhouse gas emissions, nitrogen and phosphorus emissions.

In this study, we conducted a comprehensive assessment of the nutritional and environmental impacts of commonly consumed aquatic food in China, considering 10 nutrients and multiple environmental footprints such as energy use, freshwater use, greenhouse gas emissions, and nitrogen and phosphorus emissions. By calculating nutrition and environmental impacts scores, we aimed to identify high sustainable and low sustainable species and provide dietary recommendations accordingly. Our objectives were to (1) analyze the differences and relationships between nutrition and environmental performance for each species and group, (2) develop a composite nutrition-environmental sustainability score and apply it to assess the sustainability of aquaculture in each province (excluding Taiwan, Hong Kong, and Macao), and (3) analyze various scenarios to understand how changes in aquaculture practices can affect sustainability scores for each province. Our analysis can offer valuable insights into the interplay between nutrition and multiple environmental footprints in Chinese aquaculture and provide evidence-based recommendations for sustainable dietary choices.

2. Materials and methods

2.1. Assessment of nutrient density

In this study, we categorized the species or taxa into eight groups based on their origin: marine fish, marine crustaceans, bivalves, cyprinids, freshwater fish, catfishes, tilapias, and freshwater crustaceans. To determine the nutrient densities of each species, we referred to the Food Nutrition Facts Inquiry Platform, which was established by the China Center for Disease Control and Prevention and Chinese Nutrition Society. We focused on 10 essential or micro-nutrients, including calcium, iron, zinc, phosphorus, protein, magnesium, vitamin E, selenium, monounsaturated fatty acid (M-fatty acid), and polyunsaturated fatty acids (P-fatty acids), as they are crucial for human health and most frequently deficient. If a particular species had no available nutrient density data, we substituted it with the mean nutrient densities of the same species group as a proxy.

$$Nu_{i,j} = P_i * Per_{edible,i} * ND_{i,j} \quad (1)$$

Where $Nu_{i,j}$ represents the amount of nutrition composition j for species i, P_i is the production of species i (t live weight), $per_{edible,i}$ is the percentage of edible food for species i, $ND_{i,j}$ is nutrient densities (content per 100g) of each species for nutrition composition j. The production of species come from China Fishery Statistics Yearbook. (2021).

The nutritional intensity ($NI_{i,j}$, t nutrition/t live weight production) for each nutrient of each species was calculated:

$$NI_{i,j} = \frac{Nu_{i,j}}{P_i} \quad (2)$$

Because there are multiple orders of magnitude differences in the NI of multiple nutrients, it is not possible to simply add them together. To improve the consistency, raw NI values for each nutrition composition were rescaled from 0 to 1 (Jiang et al., 2022), whereby 1 means more nutrition can be produced per unit yield and 0 means less nutrition per unit yield. The process in terms of the following formula:

$$S_{i,j} = \frac{S_{i,j} - S_{i,j,\min}}{S_{i,j,\max} - S_{i,j,\min}} \quad (3)$$

where $S_{i,j}$ denotes the standardized NI of j nutrition composition for species i, $S_{i,j,min}$ and $S_{i,j,max}$ are the minimum and maximum values of species i in the nutrition composition j, respectively.

Finally, the composite nutrition score for each species can be illustrated:

$$S_i = \sum_{j=1}^{n} S_{i,j} \quad (4)$$

2.2. Energy footprint

According to MacLeod et al. (2019), the on-farm energy use was calculated primarily for pumping water, lighting and powering vehicles, which include electricity, diesel and petrol. The energy use intensity ($EI_{E,i}$, MJ/t live weight) was calculated according to the method provided by MacLeod et al. (2019):

$$EI_{E,i} = \frac{E_{e,i} * P_i + E_{d,i} * P_i + E_{p,i} * P_i}{P_i} \quad (5)$$

Where $E_{e,i}$, $E_{d,i}$ and $E_{p,i}$ represented emissions factors for electricity, diesel and petrol applied to species i, and values of the three emissions factors extracted from MacLeod et al. (2019).

2.3. Freshwater footprint

In this study, freshwater use was estimated based on the feed-related method, which quantifies the amount of water consumed during the

production and usage of aquaculture feed (Pahlow et al., 2015). The formulas are as follows:

$$P_{feed} = P_i * Perc_{feed,i} * FCR_i \quad (6)$$

where P_{feed} is the total amount of commercial feed consumed by species i, P_i is the production (t/year) of species i and $Perc_{feed,i}$ is the percentage of commercial feed in production by species i, FCR_i is the feed conversion ratio (kg feed/kg live weight) of species i. The data of $Perc_{feed,i}$ and FCR_i come from Tang et al. (2016).

$$Feed_{i,p} = f_{i,p} * P_{feed} \quad (7)$$

where $Feed_{i,p}$ is the amount of feed ingredient p in ton/year fed to species i, and $f_{i,p}$ is the proportion of each ingredient in commercial feed applied to species i (MacLeod et al., 2019).

The formula for calculating water footprint related to commercial feed is as follows:

$$WF_{feed,i} = \sum_{p=1}^{n} Feed_{i,p} * WF_{p,i} \quad (8)$$

where $WF_{feed,i}$ is the green, blue and gray water footprint intensity of species i, and $WF_{p,i}$ is the water footprint intensity (m^3 water/t feed ingredients) of feed ingredient p, and n is the number of feed ingredients. The water footprint intensity of feed ingredients come from Pahlow et al. (2015).

The freshwater use intensity ($EI_{F,i}$, m^3/t live weight) was calculated as follows:

$$EI_{F,i} = \frac{WF_{feed,i}}{P_i} \quad (9)$$

2.4. Greenhouse gas footprint

Greenhouse gas emissions was calculated based on Xu et al. (2022) and MacLeod et al. (2019). The GHG emissions from four main aspects (i.e., feed, energy, N_2O and fertilizer) were considered. For species i, the GHG emissions intensity ($EI_{C,i}$, t CO_2e/t live weight) is calculated according to MacLeod et al. (2019):

$$EI_{C,i} = \frac{GE_{feed,i} + GE_{energy,i} + GE_{N_2O,i} + GE_{fertilizer,i}}{P_i} \quad (10)$$

where $GE_{feed,i}$, $GE_{energy,i}$, $GE_{N2O,i}$ and $GE_{fertilizer,i}$ represented GHG emissions from feed, energy, N_2O and fertilizer for species i respectively, P_i represented production of species i. The data of $Perc_{feed,i}$ and FCR_i when calculated greenhouse gas emissions also come from Tang et al. (2016).

2.5. Nitrogen footprint

Nitrogen emissions were calculated in the on-farm stage, and which was related to feed and fertilizer. Here, on-farm N emissions were estimated as the difference between the N content of feed components and fertilizer and the N contents of species. The emission intensity ($EI_{N,i}$, t N/t live weight) is calculated according to the method provided by Gephart et al. (2021):

$$EI_{N,i} = \frac{\sum_{p=1}^{n} Feed_{i,p} * N_f + N_{fertilizer} - N_i}{P_i} \quad (11)$$

Where N_f, $N_{fertilizer}$ and N_i represent the nitrogen content of feed component f, the nitrogen content of fertilizer and the nitrogen content of species i respectively. The data of N_f come from National Research Council (1982), $N_{fertilizer}$ come from Boyd et al. (2007), N_i come from Luo et al. (2018).

2.6. Phosphorus footprint

Phosphorus emissions were also calculated in the on-farm stage, which was calculated according to Huang et al. (2020).

$$Ph_{inp,i} = Ph_{hat,i} / PUE_i \quad (12)$$

Where $P_{hat,i}$ represent the phosphorus content of species i, PUE_i represent the proportion of supplementary P applied to feed species and fertilize the aquatic ecosystems that are recovered in harvested fish. The parameters of PUE for each species were driven by Huang et al. (2020).

The emission intensity ($EI_{P,i}$, t P/t live weight) for phosphorus emissions was calculated in excel, and the formula is as follows:

$$EI_{P,i} = \frac{Ph_{inp,i} - Ph_{hat,i}}{P_i} \quad (13)$$

2.7. Sustainable performance of each species

In this study, six indicators were categorized into two dimensions: nutrition provision (positive) and environmental impacts (negative). To enable a comparison of the net impacts between nutrition and environmental impacts for each species, their sustainable performance was calculated. To achieve this, the following approach was taken: Firstly, the raw values representing the intensity of each indicator for every species were arranged in descending order. Subsequently, the 49 species were categorized into 10 levels based on their raw values for each indicator. Then each interval was assigned corresponding score ranging from 1 to 10 (Li, 2014). For instance, for each indicator, the top five species with the highest raw values were assigned a score of 10, while the four species with the lowest raw values were assigned a score of 1 (Li, 2014). This approach effectively mitigated potential calculation challenges that might have arisen due to the inclusion of zero values for nutrition or environmental impacts across different species. Finally, the sustainable performance for each species was calculated as follows:

$$S_i = \frac{S_{nutrition,i}}{\left(S_{energy,i} + S_{freshwater,i} + S_{GHG,i} + S_{N,i} + S_{Ph,i}\right)/5} \quad (14)$$

where S_i (ratio between nutrient and environmental impact) is the composite sustainability score for species i, ranging from 0.1 to 10. The larger the S_i value is, the stronger the sustainability is, and *vice versa*. $S_{nutrition,i}$, $S_{energy,i}$, $S_{freshwater,i}$, $S_{C,i}$, $S_{N,i}$ and $S_{Ph,i}$ are each indicator score for species i.

2.8. Sustainable performance of Chinese aquaculture

To present the spatial changes of aquaculture's sustainability in China, we calculated the average level of sustainable development in the past 10 years (2011–2020) of each province (Taiwan, Hong Kong and Macao were not included due to the limitation of data) based on the sustainable performance for freshwater aquaculture, marine aquaculture and composite aquaculture (include freshwater and marine aquaculture). The formula for each province's freshwater aquaculture sustainability was calculated as follows:

$$S_{pf} = \frac{\sum_{i=1}^{n=28} S_i * P_i}{\sum_{i=1}^{n=28} P_i} \quad (15)$$

The calculation formula for marine aquaculture in each province was as follows:

$$S_{pm} = \frac{\sum_{i=1}^{n=21} S_i * P_i}{\sum_{i=1}^{n=21} P_i} \quad (16)$$

The formula for each province's composite aquaculture sustainability was calculated as follows:

$$S_{pc} = \frac{\sum_{i=1}^{n=49} S_i * P_i}{\sum_{i=1}^{n=49} P_i} \quad (17)$$

Where S_{pf}, S_{pm}, S_{pc} represented the freshwater, marine and composite aquaculture composite sustainability of province p respectively, the larger the values, the higher sustainability, and *vice versa*.

2.9. Scenarios for predicting the sustainability of Chinese aquaculture

Based on the sustainable performance of each species, we developed 20 scenarios by adjusting the proportion of species production with high sustainable performance ($S_i > 1$) and total production to assess the changes in the sustainability of Chinese aquaculture (marine and freshwater aquaculture) under different conditions (Table .1). Firstly, we adjusted the proportion of species production with high sustainable performance under the assumption that the total production remains constant, based on the production percentage of species with high sustainable performance (13.71%) and species with low sustainable performance (86.29%) in 2020. According to the original percentage between high sustainable performance species and low sustainable performance species, we have designed a total of five proportion gradients (−10%, −5%, +20%, +40% and +60%) for high sustainable performance species. Meanwhile, we have also designed four scenarios for total production changes (+0%, +10%, +30% and +50%). As there are many species in the two groups (high sustainable performance species and low sustainable performance species), the increased production was added according to the production proportion of the species in the same group. However, since the purpose of this scenario was to evaluate the sustainability after adjusting the species with high sustainable performance, species with a middle sustainable performance ($S_i = 1$) were not included in the calculation.

Table 1
Summary of simulation scenarios for aquaculture farming in China. All scenarios are established based on the year 2020.

Percentage of high-sustainability species ($S_i > 1$) in production	Changes in total production	Methods for increasing total Production	Scenarios code
±0%	+0%		P-T (2020 as basis)
−10%	+0%	All of the production growth was achieved by increasing the production of species with low sustainability performance ($S_i < 1$) (species with a sustainability score of less than 1); the percentage of production for low-sustainability species in the year 2020 will increase in proportion to the total production volume.	P-10%&T
−10%	+10%		P-10%& T+10%
−10%	+30%		P-10%& T+30%
−10%	+50%		P-10%& T+50%
−5%	+0%		P-5%&T
−5%	+10%		P-5%& T+10%
−5%	+30%		P-5%& T+30%
−5%	+50%		P-5%& T+50%
+20%	+0%	All of the production growth was achieved by increasing the production of species with high sustainability performance ($S_i > 1$) (species with a sustainability score of more than 1); the percentage of production for high-sustainability species in the year 2020 will increase in proportion to the total production volume.	P+20%&T
+20%	+10%		P+20%& T+10%
+20%	+30%		P+20%& T+30%
+20%	+50%		P+20%& T+50%
+40%	+0%		P+40%&T
+40%	+10%		P+40%& T+10%
+40%	+30%		P+40%& T+30%
+40%	+50%		P+40%& T+50%
+60%	+0%		P+60%&T
+60%	+10%		P+60%& T+10%
+60%	+30%		P+60%& T+30%
+60%	+50%		P+60%& T+50%

In this study, all the gathered data are sourced from literature or databases (see Table S2 for details). The calculation of nutrient content and the assessment of the magnitude of each footprint were conducted using Excel software. The emission factor method was employed for calculating each footprint. Moreover, all graphical representations of the results were generated using R software version 4.1.

3. Results

3.1. Sustainable performance of the 49 aquaculture species

Based on the assessment of composite nutrition and composite environmental impacts, certain species such as common eel, macro brachium rosenbergii, freshwater shrimp, swimming crab, and trout had the highest nutritional content, but also had high environmental impacts. Conversely, species such as tilapias, oysters, crawfish, and corbicula had the lowest nutritional content, with tilapia having the highest composite environmental impacts score. The nutritional intensity scores varied among different species groups, with freshwater crustaceans having the best score for vitamin E and bivalves performing best for iron and zinc, but having the lowest scores for M-fatty acid, phosphorus, and protein (Fig. S1). It is important to consider multiple nutrient types when assessing aquatic food. Among the five environmental impacts, bivalves and cyprinids had lower average environmental impacts scores. Marine fish and catfish had lower scores in four environmental impacts, while marine crustaceans and tilapias had only one environmental impact score lower than average (Fig. S2). The composite sustainability (S_i) was evaluated by adjusting the proportion of species production with more sustainable performance or less sustainable performance. When the nutritional score was equal to the composite environmental score, expressed as a S_i (Ratio) = 1, the positive and negative effects were offset. A higher ratio indicated a stronger sustainable performance, while a lower ratio indicated a lower sustainable performance. Mussel had the highest ratio ($S_i = 2.65$) and was the most sustainable species, while tilapia had the lowest sustainable performance ($S_i = 0.11$) compared to other species due to its low nutritional score and high environmental impact (Fig. 1, Table S1). The average ratio for marine fish, bivalves, and marine crustaceans were all above 1, with values of 1.33, 1.21, and 1.06, respectively. In contrast, the remaining five groups had an average ratio less than 1 (Fig. S3).

3.2. Aquaculture's sustainability of each province

Based on an assessment of sustainability scores and production levels for each species, the results showed that the average sustainability score was 0.52 for freshwater aquaculture, 0.84 for marine aquaculture, and 0.63 for composite aquaculture. The sustainability score of Qinghai and Tibet provinces was the highest (with scores of 1.06 and 0.75 respectively) among all provinces in freshwater aquaculture, while Hainan province had the lowest score (0.16) (Fig. 2). There were great differences in the level of sustainable development among provinces in marine aquaculture. Zhejiang province had the highest sustainability score, reaching 1.42, while Guangxi province had the lowest score, only 0.56 (Fig. 2). When considering freshwater and marine aquaculture comprehensively, Qinghai, Zhejiang, Shandong and Hebei provinces had the highest sustainable performance compared with other provinces (Fig. 2). However, when considering both freshwater and marine

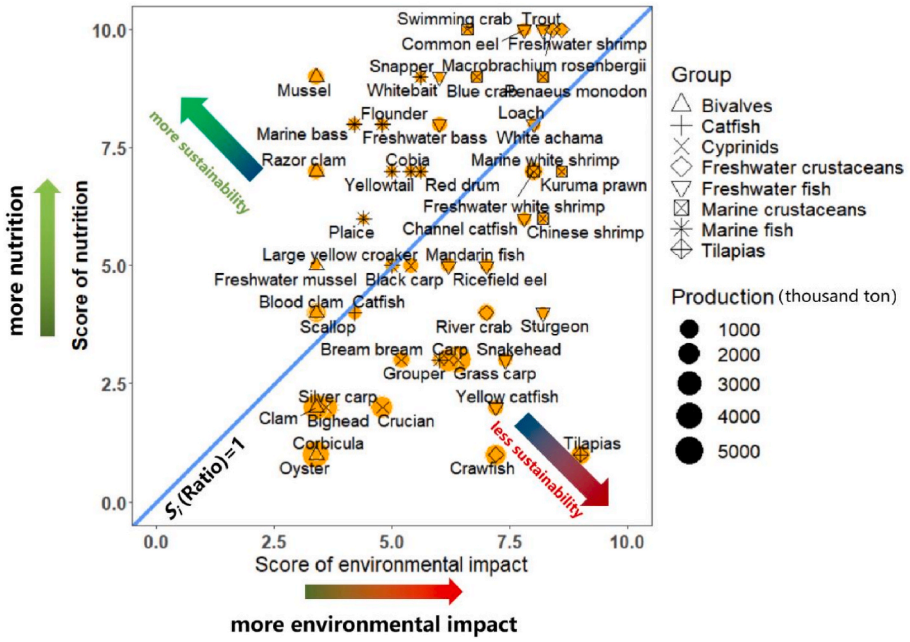


Fig. 1. Sustainable performance of each species. The blue line [S_i (Ratio) = 1] represents the score of nutrition is equal to the score of composite environmental impact. Environmental impact includes energy, freshwater use and greenhouse gas, nitrogen and phosphorus emissions. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

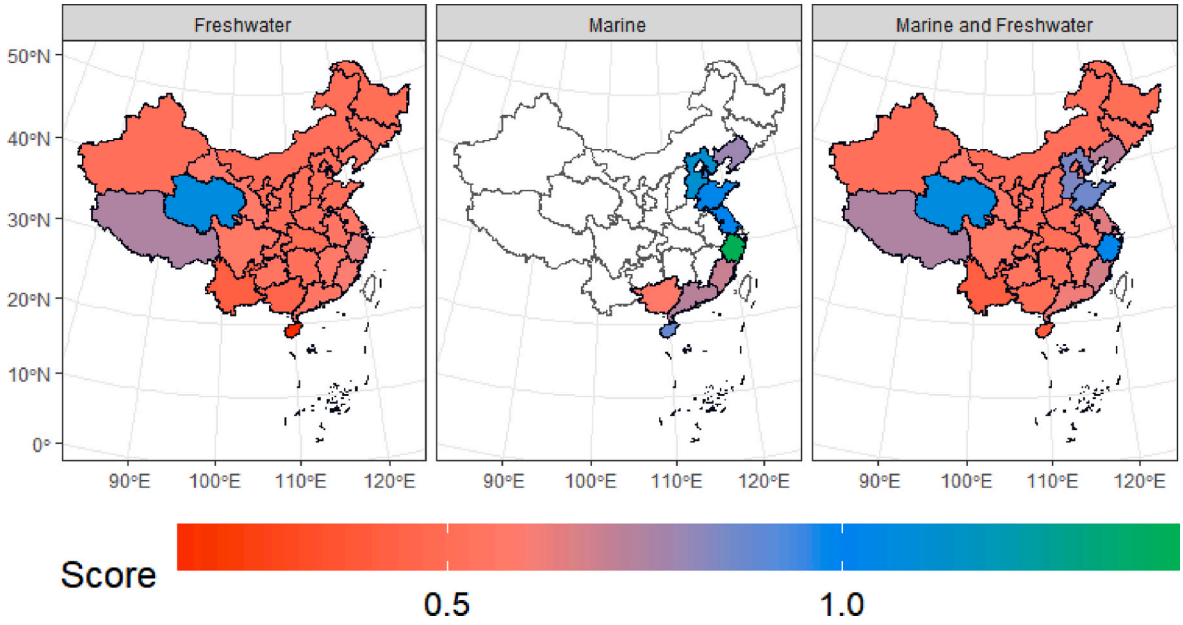


Fig. 2. Sustainable performance of aquaculture in each province. The production used the average value of 2011–2020 (there are no data in Taiwan, Hong Kong and Macao provinces, and there are no data in Shanghai province for marine aquaculture).

aquaculture, the sustainability performance of coastal provinces was reduced to varying degrees compared to only considering marine aquaculture. In particular, Hainan saw its sustainability score drop from 0.88 to 0.40 (Fig. 2).

3.3. Simulating aquaculture sustainability

When it comes to reducing the production proportion of low sustainable performance species and increasing total production by increasing the production of these species, there were not obvious changes in most provinces based on the 2020 data (Fig. 3). However, Zhejiang province stands out, as its sustainability score decreases from 1.09 in 2020 to 0.64 in the P-10%&T+50% scenario. On the other hand, when the proportion of high sustainable performance species is increased, and total production is increased by increasing their production, the sustainability score of many provinces, especially coastal provinces, increases significantly (Fig. 3). For instance, when the proportion of high sustainable performance species increases by 60% and total production increases by 50%, Zhejiang province's sustainability score jumps to 1.84. Moreover, the sustainability of inland provinces, such as Tibet and Gansu province, also improves significantly. However, increasing the total production by increasing the production of high

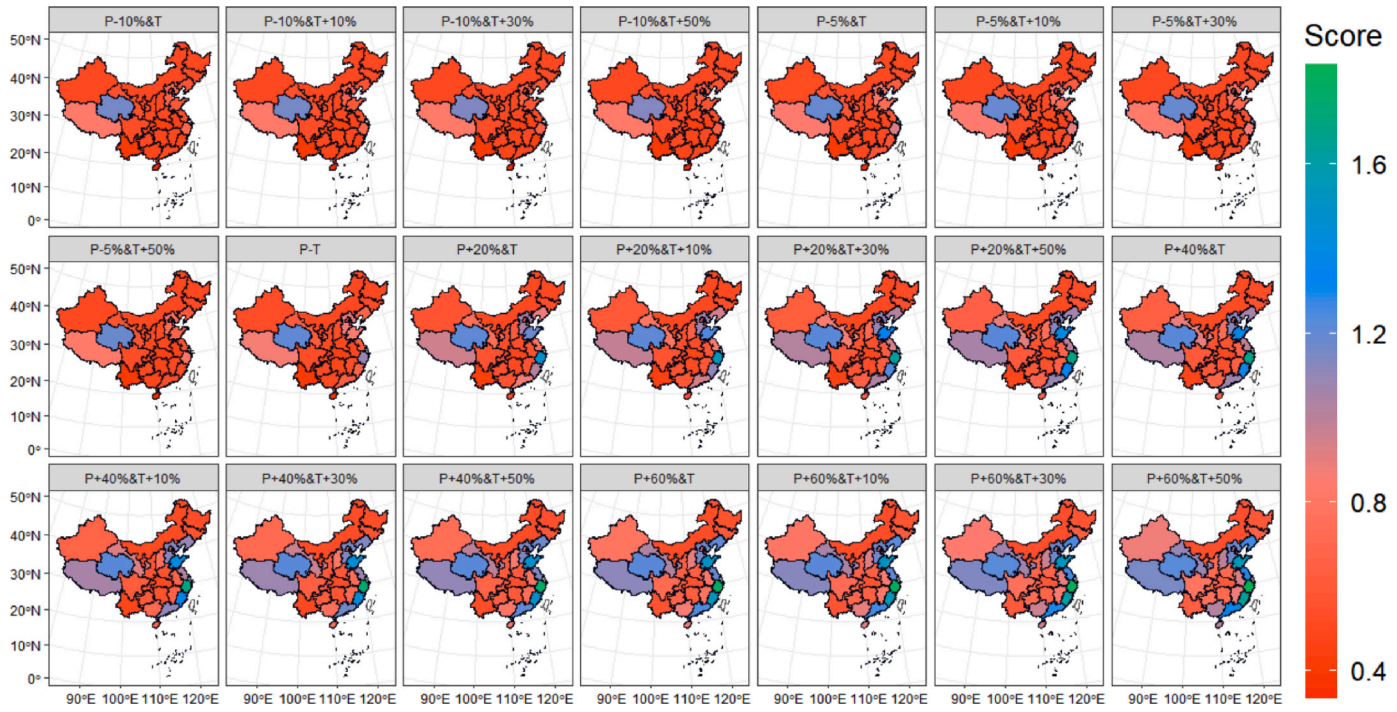


Fig. 3. Sustainable performance of Chinese aquaculture (marine and freshwater aquaculture) for each province in different scenarios. There are no data in Taiwan, Hong Kong and Macao provinces, and Shanghai province also excluded from this map due to there are no data for marine aquaculture; the title for each map can be find in Table 1.

sustainable performance species results in a smaller increase in the sustainability score compared to increasing the proportion of high sustainable performance species. For example, the sustainability score difference between P+60%&T and P+20%&T was 0.32, which is greater than the difference between P+20%&T+50% and P+20%&T, which was 0.21. Conversely, reducing the proportion of high sustainable performance species and increasing the production of low sustainable performance species will result in only a small change in the sustainability score. This is because the original production of low sustainable performance species is relatively high at 86.29%. Overall, these findings suggest that increasing the proportion of high sustainable performance species is an effective way to improve the sustainability score in most provinces, particularly coastal ones.

4. Discussion

This study is the first to analyze the sustainability of aquatic food in China based on its nutrient content and negative environmental impacts. Its aim is to provide new dietary guidelines that promote more sustainable seafood consumption in the country, by taking into account both nutrition and the various environmental impacts of aquaculture. Compared to other types of food, seafood is considered an important sector that can help minimize environmental damage while still meeting nutritional requirements (Gephart et al., 2016). Given that China has the world's largest aquaculture industry, this study can provide valuable insights into how to optimize nutrient supply and reduce environmental impacts for future aquaculture development in the country (Cao et al., 2015; Gui et al., 2018).

4.1. Sustainable performance of the 49 aquaculture species

The study found that cyprinids are a significant source of nutrition across various nutrient types, largely due to their high production in China. This can be attributed to several factors, including the species' ability to thrive in different environments and temperatures, making them a popular choice among fishermen (Orban and Wu, 2008). Additionally, cyprinids are relatively affordable compared to other seafood options, making them accessible to a wider population with economic limitations (Crona et al., 2020). Finally, the culture of cyprinid also plays a crucial role in their high production rates, which is closely tied to social factors (Osmundsen et al., 2020).

The sustainability of species currently cultured in China varies widely in terms of their environmental impacts, with significant differences between each species. As a result, it would be inaccurate to assess the sustainability of a species based on a single environmental impact (such as GHG emissions) or nutrition (such as protein) alone. However, differences in the nutritional content of each species can be used as a guide for individuals to choose suitable species in their diet when they are deficient in certain nutrients (Hallström et al., 2019). For example, individuals can consume more bivalves when they lack iron and zinc (Fig. S1). Freshwater crustaceans, which have the highest levels of calcium and vitamin E, were found to have low sustainability scores due to their high environmental impacts (MacLeod et al., 2020; Gephart et al., 2021). Therefore, when some species have high nutritional value, priority should be given to improving their environmental impacts (such as in the case of freshwater shrimp and trout).

According to the five environmental indicators, bivalves and cyprinids performed the best compared to other species groups at the species group level, with silver carp, bighead, mussel, and razor clam also performing well at species level. The difference in feed use between each species group can explain this phenomenon, as feed is the main source of environmental impacts. Aquaculture feed not only requires natural resources like freshwater and energy use but also generates significant environmental stressors like GHG, N, and P emissions through excrements, uneaten feed, or by-products (Herath and Satoh, 2015; Pahlow et al., 2015; Zhang et al., 2018). Bivalves, which are filter-feeding taxa, require little to no feed during the cultured process as they obtain their feed mainly from the surrounding environment (Froehlich et al., 2018). Furthermore, bivalves have higher nutritional intensity compared to most other species groups, making them an excellent choice for promoting a healthy diet. Silver carp and bighead carp are non-fed and unfertilized fin-fish groups (Yan et al., 2009), resulting in lower

environmental impacts compared to most other species (Fig. 1). The proportion of silver carp and bighead carp production accounts for about 35.36% of cyprinids (China Fishery Statistical Yearbook, 2020), which leads to lower environmental impacts for cyprinids. However, most cyprinidae fishes have lower nutritional intensity than most other species, so their sustainability should be assessed by combining nutrition and environmental impacts. Although marine fish is the species group that requires feed, it has higher composite scores than other species groups due to its high nutrition (with most marine fish having a ratio above 1). Thus, bivalves and marine fish are the most recommended species for a sustainable diet at the species group level. On the other hand, tilapia not only has the lowest nutritional intensity among the eight species groups but also has high environmental intensity (like high freshwater use intensity, N, and P emissions intensity), resulting in the lowest composite score (MacLeod et al., 2019). At the species level, five species or taxa have composite scores exceeding 1.6, with mussel, razor clam, marine bass, flounder and snapper being the most recommended species according to their composite scores. Therefore, they could be promoted for increased consumption and dietary advice.

4.2. Aquaculture's sustainability of each province

The differences in sustainability performance between provinces are mainly caused by the different breeding species used. For example, in Qinghai, trout is the main breeding species, while in Tibet, mandarin fish and trout are the primary species (China Fishery Statistics Yearbook, 2021). These species have relatively high sustainable performance scores, which contributes to the relatively high sustainable performance of these provinces. In contrast, since tilapias and cyprinids are the main cultured species in Hainan province and these species have low sustainable performance, the sustainability of freshwater aquaculture in Hainan province is low. In comparison with marine aquaculture, it is more urgent to improve the sustainability of freshwater aquaculture. Thus, each province can improve sustainable performance by adjusting the structure of breeding species based on their sustainable performance scores. For example, whitebait is the most sustainable species among freshwater fish, and the production ratio of this species can be appropriately increased during the breeding process for each province. Currently, the sustainable level of Chinese aquaculture is relatively low, and the proportion of high sustainable species production is also low. Therefore, the proportion of species production should be adjusted in the future, and high sustainable species should be cultivated, such as mussel, razor clam, marine bass, flounder and snapper. Especially for coastal provinces with rich marine resources, more shellfish and marine fish can be cultivated. Besides, cyprinids are the main cultured species in most provinces, so it is also crucial to improve the sustainability of cyprinids by improving the farming process.

4.3. Simulating aquaculture sustainability

Based on scenarios simulation, all provinces had improvement when the production percentage of more sustainability species increased. But coastal provinces have more marine fish and bivalves, so the sustainability of coastal provinces increases more greatly after adjusting the proportion of high sustainable performance species. Therefore, to reduce environmental impacts, it is urgent to improve the structure of aquaculture species and strengthen the cultivation of species with high sustainability, especially for coastal provinces. However, according to the results of different scenarios, increasing the production of sustainable species has a very limited effect on improving sustainability. To further improve the sustainability of aquaculture, it is not enough to adjust the structure of cultured species and further optimize the culture process. The environmental impacts of breeding species can be reduced by improving breeding methods and resource utilization rates. For example, marine fish and freshwater fish can achieve higher sustainability by improving the feeding process (Cottrell et al., 2020). The feed conversion ratio (*FCR*) is an important factor and reducing it can lead to fewer environmental impacts (Taelman et al., 2013; Dong et al., 2022). This means that less feed is used to produce the same amount of production. Changes in feed composition are also effective measures to reduce the stressors of GHG, N, and P emissions (Tacon et al., 2009; Boyd et al., 2020; Sampantamit et al., 2020). The use of more clean energy can also help reduce the environmental impacts of aquaculture, but this requires strong guidance from government departments. In addition, the continuous optimization of the farming model can also increase the farming income while reducing environmental pollution (Bergman et al., 2020).

4.4. Limitations and future directions

This study provides the first dietary advice for aquatic food in China based on both nutrition and negative impacts. However, there are some limitations to this study due to data constraints. Firstly, some parameters were calculated based on species group, such as GHG emissions intensity, which may not accurately reflect the impacts of individual species. Secondly, the equal weighting of negative impacts and nutrition, as well as the five environmental indicators, may overestimate some indicators and underestimate others. Additionally, this study only includes species cultured in China, thus excluding globally consumed species such as tuna, mussels, and wild fish. To conduct a more comprehensive analysis, future studies should include more aquatic food species. Moreover, the study only considers five environmental indicators, and additional important indicators such as land use and biodiversity footprints may need to be incorporated to identify more sustainable aquatic species (Ahmed et al., 2019; Davis et al., 2021; Gephart et al., 2021; Wilting et al., 2021).

Comprehensive advice on aquatic food should not only consider environmental impacts and nutrition but also take into account local economic levels, feeding environments (such as coastal or inland provinces and temperature), and eating habits (Béné et al., 2016; Gui et al., 2018). Furthermore, a more comprehensive method or model can be designed for analysis by considering the interdisciplinary nature of society, economy, environment, culture, trade, and breeding return rate (Hai and Speelman, 2020; Krause et al., 2020). For instance, although tilapia had the lowest sustainable performance, some breeders prefer it due to its high-density conditions, rapid growth, low mortality rate, suitability for aquaculture, stable market prices, and flexibility in feed acceptance (Maclean et al., 2002; Prabu et al., 2019). Therefore, these factors should also be included in the analysis. Additionally, collecting more accurate and complete relevant data and information by the government, social groups, or researchers can contribute to more accurate estimation, thereby enabling better evaluation of the sustainable development of aquaculture.

5. Conclusion

This study is the first step in jointly considering the nutritional benefits and environmental impacts of aquatic food as a basis for dietary advice and the development of aquaculture in China. The species used for aquaculture in China differ widely in nutritional values and environmental impacts, making them useful as a guideline for people to choose suitable species for their diets. According to sustainable performance, mussel, razor clam, marine bass, flounder and snapper are the most recommended species. In contrast, tilapia has low nutritional value and high environmental impacts, making it unsuitable for dietary advice. When freshwater and marine aquaculture are both considered, the sustainable performance of all coastal provinces is reduced to varying degrees. Therefore, adjusting the proportion of species production and improving breeding methods and resource utilization rates can significantly increase the sustainability of Chinese aquaculture. This study has limitations as it only took into account nutritional and environmental aspects, and future studies should consider additional factors

such as social, economic, and dietary factors to obtain more precise results.

CRediT authorship contribution statement

Congjun Xu: Conceptualization, Formal analysis, Data curation, Writing – original draft, Visualization. **Guohuan Su:** Conceptualization, Methodology, Formal analysis, Data curation, Writing – review & editing, Supervision, Funding acquisition. **Kangshun Zhao:** Formal analysis, Writing – review & editing. **Xiaoqi Xu:** Methodology, Writing – review & editing. **Ziqi Li:** Methodology, Formal analysis. **Min Zhang:** Conceptualization, Methodology, Writing – review & editing. **Jun Xu:** Conceptualization, Methodology, Data curation, Writing – review & editing, Project administration, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2023.138619.

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